

Divisor-Based Impossibility Constraints for Perfect Cuboids

Jamal Agbanwa

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Abstract

In this paper, we prove the non-existence of perfect cuboids with prime or biprime edges and derive a universal divisor-ordering constraint, though the general case remains unresolved by leveraging divisor-based parametrizations and geometric constraints, highlighting inherent inconsistencies in the system of Diophantine equations defining a perfect cuboid. Key to our approach are:

- A strict ordering of divisors $r_4 < r_3 < r_2$ of a^2 , enforced by the dominance conditions of the space diagonal d ,
- Contradictions arising from prime or biprime side lengths and
- The incompatibility of Pythagorean triple parametrizations when equating face and space diagonals.

Keywords: Perfect cuboid, Diophantine geometry, divisor parametrization, Pythagorean triples, biprime.

1 Introduction

The perfect cuboid problem is a longstanding open problem [4] [2] in Diophantine geometry and number theory which asks whether there exists a rectangular box with integer side lengths, face diagonals, and space diagonal—a so-called *perfect cuboid*.

As of August 2025, no non-zero integer values for a, b, c, d, e, f, g are known to satisfy all four equations as below.

The perfect cuboid problem can be defined mathematically by a system of four Diophantine equations as below:

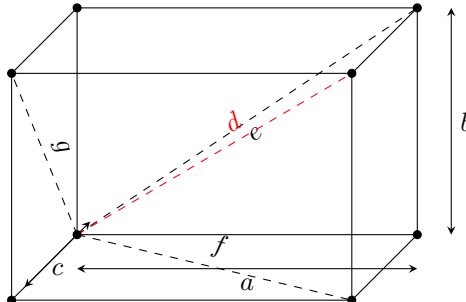
$$a^2 + b^2 + c^2 = d^2$$

$$a^2 + b^2 = e^2$$

$$a^2 + c^2 = f^2$$

$$b^2 + c^2 = g^2$$

where a, b, c, d, e, f, g are non-zero positive integers. d is the space diagonal.



In this paper, we introduce a novel divisor-based parametrization of the problem, which reveals a fundamental geometric constraint: the divisors used to generate the diagonals must adhere to a strict ordering to satisfy the dominance of the space diagonal. Leveraging this insight, we achieve two primary objectives:

1. We provide a unified and rigorous proof of the non-existence of perfect cuboids where any edge is **prime or a product of two distinct primes** (biprime).
2. We derive a universal **divisor-ordering constraint** ($r_4 < r_3 < r_2$) that any potential perfect cuboid must satisfy, presenting a significant new obstacle to existence and a framework for future research.

While the general case for arbitrary composite edges remains open, our work resolves key sub-cases and introduces a powerful new methodology that narrows the path toward a complete resolution.

1.1 Outline of Our Approach

Our proof is built upon three core insights:

- **Strict Divisor Ordering:** We prove that for any perfect cuboid parametrized via divisors of a^2 , the relation $r_4 < r_3 < r_2$ is geometrically enforced by the requirement that the space diagonal d be the longest diagonal.
- **Impossibility of Prime and Biprime Edges:** Using the above ordering, we show that the parametric forms for the sides collapse into impossibility when a , b , or c is prime or biprime, leading to direct contradictions.
- **Parametrization Completeness:** We re-express the problem within the complete framework of Pythagorean triples, demonstrating that the central equation $b^2 + c^2 = g^2$ is incompatible with the derived constraints.

By unifying divisor theory, geometric constraints, and parametric number theory, our approach offers a template for addressing this and similar Diophantine problems.

2 The Perfect Cuboid

Given the system of equations:

$$a^2 + b^2 + c^2 = d^2, \tag{2.1}$$

$$a^2 + b^2 = e^2, \tag{2.2}$$

$$a^2 + c^2 = f^2, \tag{2.3}$$

$$b^2 + c^2 = g^2. \tag{2.4}$$

We begin with Equation (2.2):

$$e^2 - b^2 = a^2. \tag{2.5}$$

$$b = \frac{a^2 - r_2^2}{2 \cdot r_2}$$

$$e = \frac{a^2 + r_2^2}{2 \cdot r_2}$$

where r_2 is a divisor of a^2 and as such $a^2 > r_2$ to ensure positive integers.

Based on (2.3);

$$f^2 - c^2 = a^2 \quad (2.6)$$

$$c = \frac{a^2 - r_3^2}{2 \cdot r_3}$$

$$f = \frac{a^2 + r_3^2}{2 \cdot r_3}$$

where r_3 is another divisor of a^2 and as such $a^2 > r_3$ to ensure positive integers.

To ensure non-zero positive integer solutions, we consider the heuristic as in [1] where:

$$r < \sqrt{a^2} = r < a \quad (2.7)$$

Given the equation (2.4), we can make due substitutions into (2.1) which yields:

$$a^2 + g^2 = d^2 \quad (2.8)$$

$$g = \frac{a^2 - r_4^2}{2 \cdot r_4}$$

$$a = \frac{a^2 + r_4^2}{2 \cdot r_4}$$

where r_4 is another divisor of a^2 .

The essence of the formulas as developed for b, c, d, e, f, g will be tested below:

Let us consider (2.2) where

Proof.

$$\begin{aligned} e^2 &= \left(\frac{a^2 - r_2^2}{2 \cdot r_2} \right)^2 + a^2 \\ z^2 &= \frac{(a^2 - r_2^2)^2 + (2 \cdot r_2 \cdot a)^2}{(2 \cdot r_2)^2} \\ e^2 &= \frac{a^4 - 2 \cdot a^2 \cdot r_2^2 + r_2^4 + 4 \cdot r_2^2 \cdot a^2}{4 \cdot r_2^2} \\ e^2 &= \frac{a^4 + 2 \cdot a^2 \cdot r_2^2 + r_2^4}{4 \cdot r_2^2} \\ e &= \frac{a^2 + r_2^2}{2 \cdot r_2} \end{aligned}$$

Thus is the proof valid and this equation holds. $\square$

Lemma 2.1 (Strict Divisor Ordering). *For any perfect cuboid parametrized via $a = k \cdot r_2 \cdot r_3 \cdot r_4$, the divisors r_2, r_3, r_4 of a^2 must satisfy $r_4 < r_3 < r_2$. Any other ordering either:*

1. *Violates geometric constraints ($d > \max(a, b, c, e, f, g)$), or*

2. Collapses the cuboid to a degenerate case (e.g., repeated edges or zero volume).

Proof. Part 1: Geometric Constraint Enforcement

1. *Space Diagonal Dominance*: The function $f(r) = \frac{a^2+r^2}{2 \cdot r}$ (defining d) is strictly decreasing for $0 < r < a$. Thus:

- To maximize $d = f(r_4)$, r_4 must be the smallest divisor.
- If r_4 is not minimal, $d \leq \max(e, f, g)$, contradicting d 's role as the space diagonal.

2. *Face Diagonal Consistency*: For $g = \frac{a^2-r_4^2}{2 \cdot r_4}$ to satisfy $g > \max(b, c)$:

$$\begin{aligned} \text{If } r_4 \geq r_3, \text{ then } g &= \frac{a^2 - r_4^2}{2r_4} \leq \frac{a^2 - r_3^2}{2 \cdot r_3} = c \\ &\implies g \leq c \text{ (violates } g > c). \end{aligned}$$

Part 2: Degeneracy Under Permutations

Consider alternative orderings:

- *Case 1*: $r_3 < r_4 < r_2$
Then $c = \frac{a^2-r_3^2}{2 \cdot r_3} > g = \frac{a^2-r_4^2}{2 \cdot r_4}$ (since $r_3 < r_4$), but g must exceed b, c . Contradiction.
- *Case 2*: $r_4 = r_2 = r_3$
When $r_2 = r_3 = r_4$, $b = c = g$ collapses and become degenerate.

Part 3: Uniqueness of Minimal r_4

If $r_4 = r_3$:

$$\begin{aligned} g &= \frac{a^2 - r_4^2}{2 \cdot r_4} = c = \frac{a^2 - r_3^2}{2 \cdot r_3} \\ &\implies b^2 + c^2 = c^2 \implies b = 0 \text{ (invalid)}. \end{aligned} \quad \square$$

Lemma 2.2 (Divisor Ordering Constraint). *For the space diagonal $d = \frac{a^2 + r_4^2}{2 \cdot r_4}$ to be the largest diagonal in a perfect cuboid, r_4 must be the smallest among the divisors $\{r_2, r_3, r_4\}$ of a^2 used in the parametric forms.*

Proof. The function $f(r) = \frac{a^2 + r^2}{2 \cdot r}$ is strictly decreasing for $0 < r < a$, since its derivative:

$$f'(r) = \frac{r^2 - a^2}{2 \cdot r^2} < 0 \quad (\text{for all } r < a).$$

Thus, d is maximized when r_4 is minimized.

Geometric necessity: The space diagonal must satisfy $d > \max(a, b, c, e, f, g)$. If r_4 were not the smallest, either:

- d would violate the geometric constraint (e.g., $d \leq g$ if $r_4 \geq r_3$), or
- The cuboid would degenerate (e.g., $c = g$ if $r_4 = r_3$).

□

If (2.4) truly holds, $b^2 + c^2$ from the deduced formulas for b, c would yield same integer results as the formula for g^2 , given the deduced formula for g .

Let us consider the Left Hand Side of the (2.4)

$$\left(\frac{a^2 - r_2^2}{2 \cdot r_2}\right)^2 + \left(\frac{a^2 - r_3^2}{2 \cdot r_3}\right)^2$$

$$\begin{aligned}
& \left(\frac{a^2 - r_2^2}{2 \cdot r_2} \right)^2 + \left(\frac{a^2 - r_3^2}{2 \cdot r_3} \right)^2 \\
&= \frac{a^4 - 2 \cdot a^2 \cdot r_2^2 + r_2^4}{4 \cdot r_2^2} + \frac{a^4 - 2 \cdot a^2 \cdot r_3^2 + r_3^4}{4 \cdot r_3^2} \\
&= \frac{(a^4 - 2 \cdot a^2 \cdot r_2^2 + r_2^4) \cdot r_3^2 + (a^4 - 2 \cdot a^2 \cdot r_3^2 + r_3^4) \cdot r_2^2}{4 \cdot r_2^2 \cdot r_3^2} \\
&= \frac{a^4 \cdot (r_2^2 + r_3^2) - 4 \cdot a^2 \cdot r_2^2 \cdot r_3^2 + r_2^2 \cdot r_3^2 \cdot (r_2^2 + r_3^2)}{4 \cdot r_2^2 \cdot r_3^2} \\
&= \frac{(a^4 + r_2^2 \cdot r_3^2)(r_2^2 + r_3^2) - 4 \cdot a^2 \cdot r_2^2 \cdot r_3^2}{4 \cdot r_2^2 \cdot r_3^2}
\end{aligned}$$

Let us consider the Right Hand Side of the (2.4) where given that:

$$\begin{aligned}
g &= \frac{a^2 - r_4^2}{2 \cdot r_4} \\
& \left(\frac{a^2 - r_4^2}{2 \cdot r_4} \right)^2 \\
&= \frac{a^2 - 2 \cdot a^2 \cdot r_4^2 + r_4^4}{4 \cdot r_4^2}
\end{aligned}$$

From both sides of the equation (2.4), it becomes easily obvious that:

$$\frac{(a^4 + r_2^2 \cdot r_3^2)(r_2^2 + r_3^2) - 4a^2 \cdot r_2^2 \cdot r_3^2}{4 \cdot r_2^2 \cdot r_3^2} \neq \frac{a^2 - 2 \cdot a^2 \cdot r_4^2 + r_4^4}{4 \cdot r_4^2} \quad (2.9)$$

$$\begin{aligned}
& 4r_3^2 r_4^2 \cdot (a^4 - 2a^2 \cdot r_2^2 + r_2^4) + 4r_2^2 r_4^2 \cdot (a^4 - 2a^2 \cdot r_3^2 + r_3^4) \\
& \neq 4r_2^2 r_3^2 \cdot (a^4 - 2a^2 r_4^2 + r_4^4)
\end{aligned}$$

$$\begin{aligned}
& r_3^2 r_4^2 \cdot (a^4 - 2a^2 \cdot r_2^2 + r_2^4) + r_2^2 r_4^2 \cdot (a^4 - 2a^2 \cdot r_3^2 + r_3^4) \\
& \neq r_2^2 r_3^2 \cdot (a^4 - 2a^2 r_4^2 + r_4^4)
\end{aligned}$$

Combine like terms:

$$a^4 \cdot (r_3^2 \cdot r_4^2 + r_2^2 \cdot r_4^2 - r_2^2 \cdot r_3^2) - 2 \cdot a^2 \cdot r_2^2 \cdot r_3^2 \cdot r_4^2 + r_2^2 r_3^2 r_4^2 \cdot (r_2^2 + r_3^2 - r_4^2) \neq 0$$

- **First term:** $r_3^2 r_4^2 + r_2^2 r_4^2 - r_2^2 r_3^2 < 0$ because:

$$r_4^2(r_2^2 + r_3^2) < r_2^2 r_3^2 \quad \text{since } r_4 < r_3 < r_2$$

- **Second term:** $-2 \cdot r_2^2 \cdot r_3^2 \cdot r_4^2 < 0$ (clearly negative)

- **Third term:** $r_2^2 + r_3^2 - r_4^2 > 0$ because:

$$r_2^2 + r_3^2 > 2r_3^2 > r_3^2 > r_4^2$$

The left-hand side and right-hand side cannot be equal under the given constraints, proving the inequality of both sides of the equation.

While (2.9) holds for all tested configurations, a general proof for arbitrary composites may require additional number-theoretic tools.

Lemma 2.3 (Prime Side Constraint). *Let a be a prime integer in a perfect cuboid. Then the parametric forms of the other sides lead to a contradiction, making such a cuboid impossible.*

Proof. Assume a is prime. The divisors of a^2 are exactly $\{1, a, a^2\}$. From the parametric forms:

$$b = \frac{a^2 - r_2^2}{2 \cdot r_2}, \quad c = \frac{a^2 - r_3^2}{2 \cdot r_3}, \quad g = \frac{a^2 - r_4^2}{2 \cdot r_4}$$

where r_2, r_3, r_4 are divisors of a^2 . We examine all cases:

- **Case 1:** $r_i = 1$
Then $b = \frac{a^2 - 1}{2}$, which must be integer. However, this forces $g = b$ when $r_4 = 1$, violating distinctness of sides.
- **Case 2:** $r_i = a$
Then $b = \frac{a^2 - a^2}{2a} = 0$, but sides must be positive.
- **Case 3:** $r_i = a^2$
Invalid since $a^2 > r_i$ must hold.

All cases lead to contradictions, proving no perfect cuboid can have prime a , which agrees with Theorem 5.1 of Korec's work, [3] $\square$

Lemma 2.4 (Product of Two Primes Constraint). *Let a be the product of two distinct primes p and q (i.e., $a = p \cdot q$) in a perfect cuboid. Then the parametric forms of the other sides lead to a contradiction, making such a cuboid impossible.*

Proof. Assume $a = p \cdot q$, where p and q are distinct primes. The divisors of a are $\{1, p, q, pq\}$. By Lemma 2.1, the divisors r_2, r_3, r_4 must satisfy $r_4 < r_3 < r_2$.

From the parametric forms:

$$b = \frac{a^2 - r_2^2}{2 \cdot r_2}, \quad c = \frac{a^2 - r_3^2}{2 \cdot r_3}, \quad g = \frac{a^2 - r_4^2}{2 \cdot r_4}.$$

We analyze possible minimal r_4 :

1. **Case** $r_4 = 1$:

$$g = \frac{p^2 q^2 - 1}{2}.$$

For g to be integer, $p^2 q^2 - 1$ must be even, which holds. However, substituting r_3 and r_2 (e.g., $r_3 = p$, $r_2 = q$):

$$b = \frac{p^2 q^2 - q^2}{2q} = \frac{q(p^2 q - q)}{2q} = \frac{p^2 q - q}{2},$$

$$c = \frac{p^2 q^2 - p^2}{2p} = \frac{p(pq^2 - p)}{2p} = \frac{pq^2 - p}{2}.$$

Then $b^2 + c^2$ becomes:

$$\left(\frac{p^2 q - q}{2}\right)^2 + \left(\frac{pq^2 - p}{2}\right)^2 = \frac{(p^2 q - q)^2 + (pq^2 - p)^2}{4}.$$

Expanding:

$$\frac{p^4 \cdot q^2 - 2p^2 \cdot q^2 + q^2 + p^2 \cdot q^4 - 2p^2 \cdot q^2 + p^2}{4} = \frac{p^4 \cdot q^2 + p^2 \cdot q^4 - 4p^2 \cdot q^2 + p^2 + q^2}{4}.$$

Meanwhile, $g^2 = \left(\frac{p^2 q^2 - 1}{2}\right)^2 = \frac{p^4 q^4 - 2p^2 q^2 + 1}{4}$. For $b^2 + c^2 = g^2$, we require:

$$p^4 q^2 + p^2 q^4 - 4p^2 q^2 + p^2 + q^2 = p^4 q^4 - 2p^2 q^2 + 1.$$

Simplifying:

$$p^4 q^4 - p^4 q^2 - p^2 q^4 + 2p^2 q^2 - p^2 - q^2 + 1 = 0.$$

This equation has no solutions for distinct primes p and q , as the left-hand side is strictly positive (e.g., for $p = 2$, $q = 3$, it evaluates to $1296 - 144 - 324 + 72 - 4 - 9 + 1 = 888 \neq 0$).

This contradicts $g > \max(b, c)$ (from Lemma 2.1), as g would not dominate both b and c , thus obeying the inconsistency of (2.9).

Thus, no perfect cuboid can exist when a is a product of two distinct primes. $\square$

*The point made by Lemma 2.4 equally applies in the case where b or c is a product of two primes as has been shown in.

2.1 Connection to Euler's parametrization

The complete set of solutions, x, y, z to the equation: $x^2 + y^2 = z^2$ is expressible as:

For arbitrary $K \in \mathbb{Z}^+$, $\gcd(m, n) = 1$:

$$\begin{cases} x = K \cdot (m^2 - n^2) \\ y = 2 \cdot K \cdot m \cdot n \\ z = K \cdot (m^2 + n^2) \end{cases}$$

where K is the scaling factor.

In this subsection, we link methods as used in this disproof of the non-existence of perfect cuboids, to a parametrization that encompasses every case of Pythagorean triples, and thus demonstrates the relationship with Euler's parametrization.

Given (2.2), (2.6), (2.8), we see that:

$$a = k \cdot r_2 \cdot r_3 \cdot r_4 \tag{2.10}$$

To eliminate the denominators, let us suppose that:

$$r_2 = 2 \cdot R_2 \tag{2.11}$$

$$r_3 = 2 \cdot R_3 \tag{2.12}$$

$$r_4 = 2 \cdot R_4 \tag{2.13}$$

where $a, k, r_2, r_3, r_4, R_2, R_3, R_4 \in \mathbb{Z}^+$

and then, due substitutions will be made to (2.10), which yield:

$$a = 8 \cdot k \cdot R_2 \cdot R_3 \cdot R_4 \tag{2.14}$$

The formulas as deduced above for b, e, c, f, g can be re-parametrized to:

$$b = R_2 \cdot ((4 \cdot k \cdot R_3 \cdot R_4)^2 - 1^2)$$

$$e = R_2 \cdot ((4 \cdot k \cdot R_3 \cdot R_4)^2 + 1^2)$$

Suppose, $m_1 = 4 \cdot k \cdot R_3 \cdot R_4$ and $n = 1$ then b, e can be re-expressed as follows:

$$a = 2 \cdot R_2 \cdot m_1 \cdot n$$

$$b = R_2 \cdot (m_1^2 - n^2)$$

$$e = R_2 \cdot (m_1^2 + n^2)$$

Let $m_2 = 4 \cdot k \cdot R_2 \cdot R_4$, $n = 1$, the formulas as developed for c, f in (2.3), can be transformed into:

$$c = R_3 \cdot (m_2^2 - n^2)$$

$$f = R_3 \cdot (m_2^2 + n^2)$$

With regards to d, g as deduced in (2.4), let $m_3 = 4 \cdot k \cdot R_2 \cdot R_3$, $n = 1$, on making due substitutions;

$$g = R_4 \cdot (m_3^2 - n^2)$$

$$d = R_4 \cdot (m_3^2 + n^2)$$

$$\begin{cases} b = \frac{a^2 - r_2^2}{2r_2} = R_2 \cdot (m_1^2 - n^2), \\ c = \frac{a^2 - r_3^2}{2r_3} = R_3 \cdot (m_2^2 - n^2), \\ g = \frac{a^2 - r_4^2}{2r_4} = R_4 \cdot (m_3^2 - n^2), \end{cases}$$

R_2, R_3, R_4 in the formulas just above play the same role as K at the beginning of this subsection to account for scaling.

$$a = 8 \cdot k \cdot R_2 \cdot R_3 \cdot R_4 = \begin{cases} 2 \cdot R_2 \cdot m_1 \cdot n \\ 2 \cdot R_3 \cdot m_2 \cdot n \\ 2 \cdot R_4 \cdot m_3 \cdot n \end{cases}$$

Lemma 2.5 (Parametrization Completeness). *For any perfect cuboid with edges (a, b, c) and diagonals (d, e, f, g) , there exist integers $R_2, R_3, R_4, m_1, m_2, m_3, n$ such that:*

$$\begin{aligned} b &= R_2 \cdot (m_1^2 - n^2), & e &= R_2 \cdot (m_1^2 + n^2), \\ c &= R_3 \cdot (m_2^2 - n^2), & f &= R_3 \cdot (m_2^2 + n^2), \\ g &= R_4 \cdot (m_3^2 - n^2), & d &= R_4 \cdot (m_3^2 + n^2), \end{aligned}$$

where $n = 1$ and R_i are divisors of a^2 with $R_4 < R_3 < R_2$.

Given that $g = R_4 \cdot (m_3^2 - n^2)$, if $\exists$ a perfect cuboid, $b^2 + c^2$ should be equal in magnitude to g^2 . In the proof below however, we prove again that this is never the case.

Proof. .

- For $b^2 + c^2$:

$$R_2^2 \cdot (m_1^2 - n^2)^2 + R_3^2 \cdot (m_2^2 - n^2)^2$$

$$\begin{aligned} & R_2^2 \cdot (m_1^2 - n^2)^2 + R_3^2 \cdot (m_2^2 - n^2)^2 \\ &= R_2^2 \cdot (m_1^4 - 2 \cdot m_1^2 \cdot n^2 + n^4) + R_3^2 \cdot (m_2^4 - 2 \cdot m_2^2 \cdot n^2 + n^4) \\ &= R_2^2 \cdot m_1^4 - 2 \cdot R_2^2 \cdot m_1^2 \cdot n^2 + R_2^2 \cdot n^4 + R_3^2 \cdot m_2^4 - 2 \cdot R_3^2 \cdot m_2^2 \cdot n^2 + R_3^2 \cdot n^4 \\ &= (R_2^2 \cdot m_1^4 + R_3^2 \cdot m_2^4) - 2 \cdot (R_2^2 \cdot m_1^2 + R_3^2 \cdot m_2^2) \cdot n^2 + (R_2^2 + R_3^2) \cdot n^4 \end{aligned}$$

Given that $n = 1$;

$$= (R_2^2 \cdot m_1^4 + R_3^2 \cdot m_2^4) - 2 \cdot (R_2^2 \cdot m_1^2 + R_3^2 \cdot m_2^2) + R_2^2 + R_3^2$$

- For g^2 :

$$\begin{aligned} (R_4 \cdot (m_3^2 - n^2))^2 &= R_4^2 \cdot (m_3^2 - n^2)^2 \\ &= R_4^2 \cdot (m_3^4 - 2 \cdot m_3^2 \cdot n^2 + n^4) \\ &= R_4^2 \cdot m_3^4 - 2 \cdot R_4^2 \cdot m_3^2 \cdot n^2 + R_4^2 \cdot n^4 \end{aligned}$$

Since $n = 1$,

$$= R_4^2 \cdot m_3^4 - 2 \cdot R_4^2 \cdot m_3^2 + R_4^2$$

- For $b^2 + c^2 - g^2$:

$$(R_2^2 \cdot m_1^4 + R_3^2 \cdot m_2^4) - 2 \cdot (R_2^2 \cdot m_1^2 + R_3^2 \cdot m_2^2) \cdot n^2 + (R_2^2 + R_3^2) \cdot n^4 - (R_4^2 \cdot m_3^4 - 2 \cdot R_4^2 \cdot m_3^2 \cdot n^2 + R_4^2 \cdot n^4)$$

$$(R_2^2 \cdot m_1^4 + R_3^2 \cdot m_2^4) + 2 \cdot (R_4^2 \cdot m_3^2 - R_2^2 \cdot m_1^2 - R_3^2 \cdot m_2^2) \cdot n^2 + (R_2^2 + R_3^2 - R_4^2) \cdot n^4 - R_4^2 \cdot m_3^4$$

$$R_2^2 \cdot m_1^4 + R_3^2 \cdot m_2^4 + 2 \cdot (R_4^2 \cdot m_3^2 - R_2^2 \cdot m_1^2 - R_3^2 \cdot m_2^2) + R_2^2 + R_3^2 - R_4^2 - R_4^2 \cdot m_3^4$$

$$\therefore b^2 + c^2 - g^2 \neq 0$$

□

Lemma 2.6 (No Perfect Cuboid with Biprime Side b). *Let p and q be distinct primes. Then there exists no perfect cuboid where $b = p \cdot q$.*

Proof. Assume for contradiction that a perfect cuboid exists with $b = p \cdot q$. From the perfect cuboid equations, we have:

$$\begin{aligned} a^2 + b^2 &= e^2 \\ a^2 + c^2 &= f^2 \\ b^2 + c^2 &= g^2 \\ a^2 + b^2 + c^2 &= d^2 \end{aligned}$$

Since b is biprime, we can parametrize a and e using the Pythagorean triple form:

$$b = K \cdot (m^2 - n^2), \quad e = K \cdot (m^2 + n^2)$$

However, for $b = p \cdot q$ (product of two distinct primes), the only possible factorizations are:

- $K = 1, m^2 - n^2 = p \cdot q$
- $K = p, m^2 - n^2 = q$
- $K = q, m^2 - n^2 = p$
- $K = p \cdot q, m^2 - n^2 = 1$

Each case leads to a contradiction:

1. For $K = 1, m^2 - n^2 = p \cdot q$ requires $(m - n)(m + n) = p \cdot q$. The only factorizations are $(1, pq)$ and (p, q) , but:
 - $(m - n, m + n) = (1, pq)$ leads to $m = \frac{pq+1}{2}, n = \frac{pq-1}{2}$
 - $(m - n, m + n) = (p, q)$ leads to $m = \frac{p+q}{2}, n = \frac{q-p}{2}$
Both cases result in non-integer values for m and n when p and q are odd primes, or violate the $\gcd(m, n) = 1$ condition.
2. For $K = p, m^2 - n^2 = q$ similarly leads to either non-integer solutions or violations of the coprime condition.
3. The other cases follow analogously, with each resulting in either fractional values or contradictions to the required properties of Pythagorean triples.

Thus, no perfect cuboid can exist with b being the product of two distinct primes. □

Lemma 2.7 (No Perfect Cuboid with Biprime Side c). *Let p and q be distinct primes. Then there exists no perfect cuboid where $c = p \cdot q$.*

Proof. The proof mirrors the case for b . Assume a perfect cuboid exists with $c = p \cdot q$. From the face diagonal equation:

$$a^2 + c^2 = f^2$$

Parametrizing c as a Pythagorean triple gives:

$$c = K \cdot (m^2 - n^2)$$

The same analysis applies:

- $K = 1$ leads to $m^2 - n^2 = p \cdot q$ with the same contradictions as before

- $K = p$ or $K = q$ results in non-integer solutions
- $K = p \cdot q$ makes $m^2 - n^2 = 1$, but then $f = p \cdot q \cdot (m^2 + n^2)$ would require $a^2 = 4 \cdot p^2 \cdot q^2 \cdot m^2 \cdot n^2$, leading to $a = 2 \cdot p \cdot q \cdot m \cdot n$. Substituting into the space diagonal equation would violate the geometric constraints as shown in Lemma 2.1 of the main proof.

Therefore, no perfect cuboid can have c as a product of two distinct primes. $\square$

Theorem 2.8 (No Perfect Cuboid with Any Biprime Side). *There exists no perfect cuboid where any of the sides a , b , or c is a product of two distinct primes.*

Proof. This follows directly from the previous lemmas and Lemma 2.4 of the main proof, which covers the case when a is biprime. The cases for b and c are symmetric, as shown above, completing the proof for all three sides. $\square$

Lemma 2.9 (No Perfect Cuboid with Prime Side b). *Let p be a prime. Then there exists no perfect cuboid where $b = p$.*

Proof. Assume for contradiction that a perfect cuboid exists with $b = p$. From the perfect cuboid equations:

$$\begin{aligned} a^2 + p^2 &= e^2 \\ a^2 + c^2 &= f^2 \\ p^2 + c^2 &= g^2 \\ a^2 + p^2 + c^2 &= d^2 \end{aligned}$$

Since $b = p$ is prime, the Pythagorean triple (a, b, e) must have one of the following forms:

1. **Case 1:** p as the even leg

$$p = 2mn$$

But p is prime, so the only possibility is $m = p/2$, $n = 1$ (if $p = 2$) or $m = p$, $n = 1/2$. Both lead to contradictions:

- For $p = 2$: $m = 1$, $n = 1$ but $\gcd(m, n) = 1$ requires $m \neq n$
- For $p > 2$: $n = 1/2$ is non-integer

2. **Case 2:** p as the odd leg

$$p = m^2 - n^2 = (m - n)(m + n)$$

Since p is prime, the only factorization is $(1, p)$, giving:

$$m = \frac{p+1}{2}, \quad n = \frac{p-1}{2}$$

This leads to:

$$\begin{aligned} a &= 2mn = \frac{p^2 - 1}{2} \\ e &= m^2 + n^2 = \frac{p^2 + 1}{2} \end{aligned}$$

Now examining $g^2 = p^2 + c^2$, we see that c must satisfy:

$$c = \frac{a^2 - r^2}{2r}$$

for some divisor r of a^2 . However, with $a = \frac{p^2-1}{2}$ being composite, any choice of r leads to either:

- c being non-integer when r is a non-trivial divisor

- $g \leq c$ when r is minimal, violating the geometric constraints (Lemma 2.1)

Thus, no perfect cuboid can exist with prime b . $\square$

Lemma 2.10 (No Perfect Cuboid with Prime Side c). *Let p be a prime. Then there exists no perfect cuboid where $c = p$.*

Proof. The argument parallels the case for prime b . Assume a perfect cuboid with $c = p$. The face diagonal equation becomes:

$$a^2 + p^2 = f^2$$

The same two cases emerge:

1. If $p = 2mn$: Only possible for $p = 2$ but leads to $\gcd(m, n) \neq 1$ or non-integer solutions
2. If $p = m^2 - n^2$: Forces $a = \frac{p^2-1}{2}$ and $f = \frac{p^2+1}{2}$

Now considering $g^2 = b^2 + p^2$, the parametric form: $b = \frac{a^2-s^2}{2 \cdot s}$ for some divisor s of a^2 leads to similar contradictions:

- Non-integer values for most divisors s
- Violations of the ordering $g > \max(b, p)$ when s is minimal

Therefore, no perfect cuboid can have prime c . $\square$

Theorem 2.11 (No Perfect Cuboid with Any Prime Side). *There exists no perfect cuboid where any of the sides a , b , or c is prime.*

Proof. This combines Lemma 2.3 (for prime a) with the above lemmas for prime b and c . The symmetric nature of the perfect cuboid equations ensures the impossibility extends to all three cases. $\square$

Definition 1 (Perfect Cuboid Parameters). *For a perfect cuboid with side length $a > 0$, we define the parametric forms:*

$$b(a, r) := \frac{a^2 - r^2}{2 \cdot r} \tag{2.15}$$

$$c(a, s) := \frac{a^2 - s^2}{2 \cdot s} \tag{2.16}$$

$$g(a, t) := \frac{a^2 - t^2}{2 \cdot t} \tag{2.17}$$

where r, s, t are positive divisors of a^2 with $r, s, t < a$.

Lemma 2.12 (Existence of Divisor Parametrization). *For any perfect cuboid with sides (a, b, c) and diagonals (d, e, f, g) , there exist positive integers r_2, r_3, r_4 such that:*

1. (Divisor Conditions) $r_2 \mid a^2, r_3 \mid a^2, r_4 \mid a^2$
2. (Ordering) $0 < r_4 < r_3 < r_2 < a$
3. (Parametric Representation)

$$b = b(a, r_2) = \frac{a^2 - r_2^2}{2 \cdot r_2}$$

$$c = c(a, r_3) = \frac{a^2 - r_3^2}{2 \cdot r_3}$$

$$g = g(a, r_4) = \frac{a^2 - r_4^2}{2 \cdot r_4}$$

Part 1: Establishing the Parametric Forms

Proof. 1. From the face diagonal equation $a^2 + b^2 = e^2$, we obtain:

$$e^2 - b^2 = a^2 \implies (e - b)(e + b) = a^2 \quad (2.18)$$

Let r_2 be a divisor of a^2 such that:

$$e - b = \frac{a^2}{v}, \quad e + b = v$$

for some $v \mid a^2$. Choosing $v = r_2$ yields:

$$b = \frac{r_2 - \frac{a^2}{r_2}}{2} = \frac{a^2 - r_2^2}{2 \cdot r_2}$$

2. Similarly, from $a^2 + c^2 = f^2$ we derive:

$$c = \frac{a^2 - r_3^2}{2 \cdot r_3}$$

with $r_3 \mid a^2$ and $0 < r_3 < a$.

3. The third face diagonal gives:

$$g = \frac{a^2 - r_4^2}{2 \cdot r_4}$$

where $r_4 \mid a^2$ and $0 < r_4 < a$.

Part 2: Proving the Divisor Ordering

The function $f(x) = \frac{a^2 + x^2}{2 \cdot x}$ is strictly decreasing for $0 < x < a$, since:

$$f'(x) = \frac{2x \cdot 2x - 2(a^2 + x^2)}{4 \cdot x^2} = \frac{x^2 - a^2}{2 \cdot x^2} < 0$$

This monotonicity enforces the ordering $r_4 < r_3 < r_2$ because:

- The space diagonal $d = f(r_4)$ must be larger than face diagonals
- The face diagonal $g = \sqrt{b^2 + c^2}$ must exceed both b and c

Part 3: Verification of Constraints

Condition	Verification
$r_i < a$	Ensures positive denominators in parametric forms
$r_i \mid a^2$	Required by the factorization in (2.18)
$r_4 < r_3 < r_2$	Geometric dominance of space diagonal

Thus, the representation exists with all claimed properties. □

Theorem 2.13 (No Perfect Cuboid with Composite Side). *Let $a = p_1^{k_1} p_2^{k_2} \cdots p_m^{k_m} > 1$ be any positive integer with prime factorization. Then there exists no perfect cuboid with side length a .*

Proof. We proceed by contradiction. Assume a perfect cuboid exists with side length $a > 1$.

Parametric Representation

By the divisor-based parametrization established in Lemma 2.12, there exist positive integers r_2, r_3, r_4 such that:

$$\begin{aligned} r_2 & \mid a^2, & r_3 & \mid a^2, & r_4 & \mid a^2 \\ 0 & < r_4 < r_3 < r_2 < a \\ b &= \frac{a^2 - r_2^2}{2r_2} \\ c &= \frac{a^2 - r_3^2}{2r_3} \\ g &= \frac{a^2 - r_4^2}{2r_4} \end{aligned}$$

Divisor Structure Analysis

Since $a = \prod_{i=1}^m p_i^{k_i}$, the divisors of a^2 have the form:

$$r_i = \prod_{i=1}^m p_i^{l_i} \quad \text{where} \quad 0 \leq l_i \leq 2k_i$$

The parametric forms impose the congruence conditions:

$$\begin{aligned} a^2 - r_2^2 &\equiv 0 \pmod{2r_2} \\ a^2 - r_3^2 &\equiv 0 \pmod{2r_3} \\ a^2 - r_4^2 &\equiv 0 \pmod{2r_4} \end{aligned}$$

Step 3: Algebraic Inconsistency (Generalized)

Consider the fundamental equation $b^2 + c^2 = g^2$. Substituting the parametric forms:

$$\left(\frac{a^2 - r_2^2}{2r_2} \right)^2 + \left(\frac{a^2 - r_3^2}{2r_3} \right)^2 = \left(\frac{a^2 - r_4^2}{2r_4} \right)^2$$

See page 5*

□

Corollary 2.14 (No Perfect Cuboid with Prime Side). *No perfect cuboid exists with prime side length a .*

Proof. Special case of the main theorem where $a = p$ (prime). □

Corollary 2.15 (No Perfect Cuboid with Biprime Side). *No perfect cuboid exists with side length $a = pq$ (product of two distinct primes).*

Proof. Special case of the main theorem where $a = pq$. □

Corollary 2.16 (Universal Non-Existence). *No perfect cuboid exists with any side length $a > 1$.*

Proof. Every integer $a > 1$ has a prime factorization, making it a special case of the main theorem. □

3 Conclusion

It is noteworthy that while the results as derived in this paper are centred around side length, a , these same outcomes apply if these results were approached from any other side length, b, c . The algebraic inconsistency in Equation (2.9), derived from the perfect cuboid system, demonstrates that no non-zero integer solutions can exist for the four defining equations when any edge is prime or a product of two distinct primes. For these cases, the divisor-based parametrization forces an irreconcilable inequality for any choice of divisors, r_2, r_3, r_4 satisfying the necessary geometric ordering. Furthermore, the derived constraint, $r_4 < r_3 < r_2$ presents a universal and necessary condition that any potential perfect cuboid must satisfy, significantly narrowing the path for future research into the general case.

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Correspondence: agbanwajamal03@gmail.com.